

1. Executive Summary

1.1 What We Are Building

Xenios is the **first AI coaching intelligence platform** that transforms how personal trainers and health coaches operate. In six weeks, we will deliver a production-ready MVP that combines automated data extraction from health documents, real-time session recording with AI-powered transcription, and intelligent coaching analytics—eliminating 10+ hours of administrative work per week while making trainers measurably better at their craft.

The platform integrates three core capabilities into a unified system: (1) an intelligent data extraction engine that automatically processes InBody scans, lab results, wearable exports, and nutrition logs into structured, actionable insights; (2) meeting intelligence that captures in-person training sessions via audio recording, transcribes conversations, and generates comprehensive session summaries in under 90 seconds; and (3) coaching performance analytics that provide trainers with objective feedback on their communication patterns, cueing effectiveness, and client engagement trends.

1.2 Who It Is For

Primary Users: Elite personal trainers and health coaches managing 20-200+ clients who want to scale their practice without sacrificing service quality. These professionals currently spend 15-20 minutes after each session manually documenting workouts, form cues, and client conversations—time that could be spent training additional clients or improving their craft.

Secondary Users: Clients of these coaches, who receive personalized session recaps, progress tracking, and a more connected coaching experience. The platform enhances the coach-client relationship by ensuring nothing discussed in sessions is forgotten and every milestone is celebrated.

1.3 Core Problems Solved

- **Documentation Burden:** Trainers spend 15-20 minutes per session on manual note-taking. With Xenios, this drops to under 2 minutes of

review time—a 90% reduction that saves 5-10 hours weekly for full-time trainers.

- **Fragmented Health Data:** Client information lives across multiple apps, PDFs, and spreadsheets. Xenios consolidates everything into a single source of truth with AI-generated insights.
- **No Coaching Feedback Loop:** Trainers receive zero systematic feedback on their coaching effectiveness. Xenios provides objective analytics on communication patterns, cueing frequency, and client sentiment trends.
- **Client Churn Blindspots:** Coaches often don't realize a client is disengaging until they've already left. Xenios detects early warning signals and prompts proactive intervention.

1.4 Success Metrics

The MVP will be considered successful when it achieves the following criteria:

Metric	Target
Time saved per session	13-18 minutes (vs. 15-20 min manual)
Session summary generation time	< 90 seconds end-to-end
Workout extraction accuracy	90%+ on exercises, sets, reps, weights
Transcription accuracy (gym environment)	≥ 85% with quality headset mic
Coach satisfaction (NPS)	≥ 50 among pilot coaches
System uptime	99.5%+ availability

The "Wow" Moment: A trainer finishes a 45-minute session, taps "Complete," and within 60 seconds receives a fully structured summary with exercises, form cues, consultation notes, and a draft message to send their client. What previously took 15-20 minutes of tedious typing is done before they can refill their water bottle.

2. Real-World Use Cases

The following scenarios demonstrate how Xenios solves real problems for personal trainers in their daily workflow.

2.1 Use Case: Complete In-Person Training Session

Context: Sarah is a 34-year-old client working with Coach Mike. She's 12 sessions into a 16-week program with a goal to lose 15 lbs before her wedding in 8 weeks. Last session, she mentioned lower back tightness during deadlifts.

Pre-Session (Automated Preparation)

Mike arrives at the gym and opens the Xenios mobile app. He immediately sees Sarah's session scheduled with a quick prep brief that includes: her last session notes (lower back tightness during deadlifts), current progress (squat increased to 125 lbs, ready for 135?), goal status (9 lbs lost, 6 lbs remaining in 3 weeks), and personal context (son has soccer tournament this weekend).

Pain Point Addressed: Without Xenios, Mike would need to scroll through text messages, check separate apps for workout history, and try to remember personal details about Sarah's life—often forgetting important context that builds client relationships.

During Session (Seamless Capture)

Mike taps "Start Session" and puts his phone in his pocket. He wears a fitness-grade Bluetooth headset mic (recommended: Samson Airline 77 or similar). For the next 45 minutes, he coaches Sarah exactly as he normally would—no behavior change required. The app quietly records the session audio in the background.

During the session, natural coaching occurs: Mike cues "chest up" and "drive through heels" during squats. Sarah mentions wedding stress and asks if she's losing weight fast enough. Mike provides encouragement and references her actual progress data. All of this conversation is captured.

Pain Point Addressed: Current solutions require trainers to stop mid-session to log exercises, breaking the coaching flow. Xenios captures everything passively—the trainer's full attention stays on the client.

Post-Session (The Magic)

Mike taps "Complete Session." Within 60 seconds, a structured summary appears on his phone containing: workout completed (exercises with exact weights recorded), form cues extracted ("chest up" mentioned 4 times during squats), consultation notes (wedding stress, reassurance provided, sleep

improving to 7 hours), personal connection reminders (follow up about son's tournament on Monday), and a draft personalized message ready to send to Sarah.

Mike reviews the summary in 45 seconds, makes one small edit (adds a fire emoji to the message), and taps "Send to Sarah." Total post-session time: under 2 minutes.

Pain Point Addressed: The 15-20 minutes Mike previously spent typing session notes is eliminated. He can immediately move to his next client or take a genuine break.

Client Experience

Sarah receives a push notification 5 minutes after leaving the gym. She opens the Xenios client app and sees her session recap: workout completed with weight progression highlighted (135 lbs on squats—up 10 lbs!), progress toward her goal (9 lbs down, on track), Mike's personalized message celebrating her wins, and clear homework items for the week. Sarah feels incredibly supported and "seen" by her coach—a level of personalization that builds loyalty and drives referrals.

2.2 Use Case: Coaching Performance Analytics

Context: Mike has been using Xenios for 12 weeks. He has 35 active clients and has conducted 420 sessions through the platform.

Monthly Analytics Review

Mike logs into the web dashboard and navigates to "My Coaching Analytics." He discovers objective data about his coaching style that he never had access to before: his talk-time ratio is 58% coach / 42% client (benchmark: elite trainers average 55/45); his most common form cue is "chest up" (mentioned 347 times this month); his clients' positive sentiment is 84% (up 6% from last month); his average message response time is 6.2 hours (top performers: 3.8 hours).

The platform provides actionable coaching tips: "Try varying your cueing language—alternatives like 'proud chest' or 'lead with sternum' can accelerate

motor learning by 15%." Mike also sees that his client retention rate improved from 87% to 94% since he started using the platform.

Pain Point Addressed: Trainers operate in a feedback vacuum—they have no systematic way to know if they're improving. Xenios provides the equivalent of a coaching mentor analyzing every session.

2.3 Use Case: Risk Detection Prevents Client Churn

Context: Mike has 35 clients. One is about to ghost. He doesn't know it yet.

Early Warning Detection

Mike opens the Xenios app Tuesday morning and sees a red notification badge: "3 clients showing disengagement patterns." He clicks into the Client Risk Dashboard and sees James Rodriguez flagged with an 87/100 risk score (High Risk).

The system explains why: session completion dropped from 100% to 75% over the past two weeks; message response time increased from 2 hours to 18 hours; sentiment analysis shows positive language down 42%; and James mentioned being "busy" three times in recent sessions. Pattern matching indicates 87% similarity to clients who cancelled within 2 weeks.

Proactive Intervention

Xenios provides recommended actions and pre-written conversation starters. Mike calls James that morning. James reveals he started a new job two weeks ago and morning workouts no longer work—he was about to cancel. Mike immediately offers evening slots and suggests reducing volume by 20% while James adjusts. James stays.

Pain Point Addressed: Client churn often happens silently. By the time a trainer notices, the client has mentally moved on. Research shows 70% retention when coaches reach out proactively versus 30% without intervention. This single saved client represents \$3,000+ in annual revenue.

2.4 Use Case: Data Extraction and Insights

Context: A new client, Marcus, just completed his onboarding assessment. He's uploaded an InBody body composition scan, recent bloodwork from his doctor, and connected his WHOOP wearable.

Automated Data Processing

Within 60 seconds of upload, Xenios automatically extracts structured data from all sources: from the InBody PDF, it pulls weight (185.4 lbs), body fat percentage (22.3%), skeletal muscle mass (78.2 lbs), and basal metabolic rate (1,847 kcal). From the lab results CSV, it extracts metabolic markers (fasting glucose: 98 mg/dL, flagged as borderline high), lipid panel (LDL: 142 mg/dL, flagged as elevated), and testosterone levels. From WHOOP, it aggregates 7/14/30-day averages for HRV, sleep duration, recovery scores, and strain levels.

The AI generates draft insight cards: "Marcus's fasting glucose (98 mg/dL) is borderline high. Combined with elevated LDL, consider discussing nutrition strategies focused on reducing refined carbohydrates." These insights appear in a coach approval queue—they are NOT shared with the client until Mike reviews and approves them.

Pain Point Addressed: Coaches currently spend 30-60 minutes manually reviewing and synthesizing client health data. Xenios reduces this to a 2-minute review of pre-analyzed, contextualized insights with full audit trail of all AI-generated content.

3. Technical Architecture

The Xenios MVP follows a **monolithic architecture** optimized for a small engineering team building in 6 weeks. This approach eliminates operational complexity while maintaining clear scaling paths for post-MVP growth.

3.1 System Overview

The system consists of five primary layers:

- **Mobile Application (React Native):** Cross-platform app for session recording, client management, and real-time notifications. Handles Bluetooth headset mic integration, background audio recording, and offline-first data sync.

- **Web Application (Next.js):** Dashboard for analytics, detailed session review, and administrative functions. Provides rich data visualization and bulk operations not suitable for mobile.
- **Backend API (Node.js/Express or Next.js API Routes):** Unified API layer handling authentication, business logic, and orchestration of AI services. Includes RBAC enforcement and comprehensive audit logging.
- **Background Workers (pg-boss):** PostgreSQL-native job queue for async processing: transcription jobs, document extraction, insight generation, analytics aggregation, and risk detection.
- **External Services:** Deepgram for ASR/transcription, IBM Docling for document extraction, Claude/GPT-4 for NLP and summarization, S3/R2 for file storage.

3.2 Component Communication

All client applications communicate with the backend via HTTPS REST APIs. Real-time updates (transcription streaming, notification delivery) use WebSocket connections. The backend orchestrates external service calls and manages job queues.

Authentication Flow: JWT-based authentication with refresh tokens. Mobile app stores tokens securely in device keychain. All API requests include authorization headers validated against PostgreSQL-backed session store.

Data Flow for Session Recording: (1) Mobile captures audio via Bluetooth headset → (2) Audio uploaded to S3 with presigned URL → (3) Upload completion triggers transcription job via pg-boss → (4) Transcription results stored in PostgreSQL → (5) Classification and summarization jobs chain automatically → (6) Push notification sent to coach when summary ready.

3.3 External Services and Integrations

Service	Provider	Purpose
Speech-to-Text	Deepgram Nova-2	Real-time transcription with speaker diarization
Document Extraction	IBM Docling (self-hosted)	PDF parsing for InBody scans, lab results

LLM Inference	Claude 3.5 Sonnet / GPT-4o	Summarization, classification, entity extraction
File Storage	AWS S3 or Cloudflare R2	Audio files, documents, exports
Database	PostgreSQL (Supabase/Neon)	Primary data store with RLS
Job Queue	pg-boss	Async job processing (PostgreSQL-native)
Push Notifications	Firebase Cloud Messaging	Mobile push notifications

3.4 Technology Stack Decisions

Document Extraction: IBM Docling (Recommended)

After evaluating Azure Document Intelligence, AWS Textract, Google Document AI, and IBM Docling, we recommend **IBM Docling** as the primary document extraction engine for MVP.

Technology	Cost/1K Pages	Table Accuracy	Setup Time	Recommendation
IBM Docling	\$0 (self-hosted)	97% TEDS	2-4 hours	PRIMARY - Best value
Azure Doc Intelligence	\$30	~95%	1-2 days	BACKUP - If Docling fails
AWS Textract	\$15-70	90-95%	1-2 days	Not recommended
Google Document AI	\$30-65	~94%	2-3 days	Not recommended

Rationale: Docling achieves 97% accuracy on table extraction (TEDS benchmark), matching or exceeding commercial alternatives. For InBody PDFs with structured tables, this is more than sufficient. The September 2025 release of Granite-Docling-258M adds vision-language capabilities while maintaining the same integration pattern.

Important Note: 50%+ of Xenios's document types don't require OCR at all. Wearable exports (Garmin, WHOOP, Apple Health) come as JSON/CSV/XML—direct parsing with standard libraries costs nothing and guarantees 100% accuracy. Reserve document intelligence for PDFs only.

Transcription: Deepgram Nova-2 (Recommended)

For gym environment recording with speaker diarization, **Deepgram Nova-2** provides the optimal combination of streaming latency (<300ms), free speaker diarization, noise robustness, and HIPAA compliance.

Service	Streaming	Cost/Hour	Diarization	HIPAA BAA
Deepgram Nova-2	Yes (<300ms)	\$0.26-0.35	Free	Yes - RECOMMENDED
AssemblyAI	Yes (<300ms)	\$0.15	Included	Yes - Backup option
OpenAI Whisper	No (batch only)	\$0.36	None	NO - Not recommended

Critical Note on Whisper: OpenAI Whisper is explicitly NOT recommended despite its popularity. It lacks native streaming, has no built-in speaker diarization, exhibits hallucination issues with noisy audio, and does not offer HIPAA BAAs. The 25MB file limit also requires chunking logic.

Hardware Requirement: Quality headset microphones for trainers are essential. Recommended options: Samson Airline 77 fitness headset (~\$200) or SYNCO G2 Max wireless lavalier (~\$150). Close-mic placement (2-6 inches from mouth) transforms unusable gym audio into clean transcription input.

3.5 Major Technical Constraints

- **Audio Quality Dependency:** Transcription accuracy is highly dependent on microphone quality and placement. Phone microphones in gym environments yield 60-70% accuracy; fitness headset mics yield 85-90%.
- **Processing Latency:** The 60-90 second target for session summary generation requires efficient pipeline orchestration. Transcription of 45-minute audio takes ~4-5 minutes; classification and summarization add ~30 seconds.
- **Mobile Background Processing:** iOS and Android impose strict limits on background audio recording. The app must maintain an active foreground service (Android) or audio session (iOS) throughout the recording.
- **Data Sovereignty:** All PHI must be processed by HIPAA-compliant vendors with signed BAAs. Audio files are deleted within 24 hours; only transcripts are retained long-term.

4. Enhanced Data Model

The data model is designed around three core principles: (1) normalized structure for query performance and data integrity; (2) comprehensive audit logging for compliance; and (3) flexible JSONB fields only where schema evolution is genuinely needed.

4.1 Core Entities

Users, Coaches, and Clients

The system uses a unified **users** table with role-based differentiation. A user can be a coach, a client, or both (some trainers hire their own coaches). The **coach_client_relationships** table manages the many-to-many relationship between coaches and their clients.

Entity	Key Fields	Purpose
users	id, email, role, created_at	All system users (coaches and clients)
coach_profiles	user_id, business_name, certifications	Extended profile data for coaches
client_profiles	user_id, goals, program_start_date	Extended profile data for clients
coach_client_relationships	coach_id, client_id, status, started_at	Links coaches to their clients

Sessions and Transcripts

The **sessions** table is the central entity for training sessions, linking to transcripts, workout logs, and generated summaries. Each session has a processing pipeline state machine tracking its progress through transcription, classification, and summarization.

Entity	Key Fields	Purpose
sessions	coach_id, client_id, started_at, processing_status, summary (JSONB)	Training session metadata and generated summary
transcript_segments	session_id, speaker, start_ms, end_ms, text, intent, confidence	Individual utterances with AI classification
workout_exercises	session_id, exercise_name, sets, reps, weight, form_cues_given	Extracted workout data from sessions
form_cues_tracking	session_id, coach_id, client_id, cue_text, exercise_name, cue_count	Coaching cue analytics aggregation

Health Data and Measurements

Health data is stored in **normalized tables** rather than JSONB to enable efficient queries and proper indexing. The **artifacts** table stores uploaded files; the **measurements** table stores extracted values.

Entity	Key Fields	Purpose
artifacts	client_id, file_type, file_url, source, extracted_at	Uploaded documents (InBody, labs, etc.)
measurements	client_id, metric_type, value, unit, recorded_at, artifact_id	Extracted health metrics (normalized)
insight_cards	client_id, insight_type, content, status, approved_by, approved_at	AI-generated insights with approval workflow
wearable_summaries	client_id, period_start, period_end, avg_hrv, avg_sleep, metrics (JSONB)	Aggregated wearable data (7/14/30-day)

Analytics and Risk Detection

Entity	Key Fields	Purpose
coaching_analytics	coach_id, period_start, period_end, talk_time_pct, avg_cues_per_session	Aggregated coaching performance metrics
client_risk_scores	client_id, risk_score, risk_level, factors (JSONB), recommended_actions	Churn prediction with intervention suggestions
events_audit	user_id, action, entity_type, entity_id, timestamp, metadata (JSONB)	Append-only audit log for compliance

4.2 Entity Relationships

- **Users → Sessions:** A coach has many sessions; a client has many sessions. Sessions link both via foreign keys.
- **Sessions → Transcript Segments:** One session has many transcript segments (one per utterance). Cascade delete on session removal.
- **Sessions → Workout Exercises:** One session has many exercises. Each exercise can link back to specific transcript_segment_ids.
- **Clients → Measurements:** One client has many measurements. Measurements optionally link to the artifact they were extracted from.
- **Insight Cards → Approvals:** Each insight card has approval state. Only approved insights are visible to clients.

4.3 Data Rules and Constraints

- **Row-Level Security (RLS):** Coaches can only access their own clients' data. RLS policies enforce this at the database level, preventing application bugs from leaking data.
- **Soft Deletes:** User accounts and sessions use soft deletion (`deleted_at` timestamp) rather than hard deletion for audit trail preservation.
- **Audio Retention:** Audio files are automatically deleted 24 hours after transcription completion. Transcripts are retained indefinitely. Coaches can opt-in to extended audio retention if explicitly required.
- **Insight Approval Gate:** AI-generated insight cards have status = 'draft' by default. They MUST be approved by a coach (status = 'approved') before becoming visible to clients. This prevents AI hallucinations from reaching end users.
- **Audit Log Immutability:** The `events_audit` table is append-only. No UPDATE or DELETE operations are permitted. This provides tamper-evident compliance logging.

5. 6-Week Development Timeline

The following timeline assumes a team of 2-3 engineers working full-time. The CTO will assign specific responsibilities based on team strengths. Milestones are structured to deliver demonstrable progress every week.

Week 1: Foundation and Infrastructure

Goal: Core infrastructure operational, basic authentication working, database schema deployed.

1. PostgreSQL database setup with Supabase or Neon (all tables created, RLS policies active)
2. Authentication system with JWT tokens and refresh token rotation
3. CI/CD pipeline configured (GitHub Actions → staging environment)
4. pg-boss job queue infrastructure operational
5. S3/R2 storage configured with presigned URL generation
6. Mobile app skeleton (React Native): auth flow, navigation structure
7. Web dashboard skeleton (Next.js): login, basic layout

Week 1 Demo: User can log into mobile app and web dashboard. Database queries return test data. File upload to S3 works.

Week 2: Data Extraction Pipeline

Goal: Upload health documents → automatic extraction → data visible in UI.

1. IBM Docling integration for PDF extraction (InBody scans, lab results)
2. CSV/JSON parsers for wearable exports (Garmin, WHOOP, Apple Health)
3. Async extraction job workers (artifact upload → trigger extraction)
4. Client profile page showing extracted measurements
5. Basic insight generation (flag out-of-range lab values)
6. Coach approval queue UI (draft insights list with approve/reject buttons)

Week 2 Demo: Upload InBody PDF → within 60 seconds, weight and body fat appear in client profile. Upload Oura CSV → HRV averages calculated and displayed.

Week 3: Audio Recording and Transcription

Goal: Record sessions on mobile → upload → receive transcription.

1. Mobile session recorder with Bluetooth headset mic support
2. Background audio recording (iOS AVAudioSession, Android Foreground Service)
3. Audio upload flow with progress indicator and offline queuing
4. Deepgram integration for batch transcription with speaker diarization
5. Transcript segment storage (speaker, timestamps, text, confidence)
6. Basic transcript viewer in web dashboard

Week 3 Demo: Record 10-minute test session on phone → upload → view transcript with coach/client speaker labels in web dashboard.

Week 4: AI Classification and Summarization

Goal: Transcripts → classified segments → structured session summaries.

1. Intent classification pipeline (workout_instruction, form_cue, consultation, off_topic)
2. Entity extraction for workout data (exercise names, sets, reps, weights)
3. Form cue extraction and counting
4. Session summary generation (LLM-powered, structured output)

5. Summary review and editing interface
6. Pre-session brief screen (last session context, suggested focus)

Week 4 Demo: Complete 45-minute session recording → tap Complete → within 90 seconds, structured summary appears with exercises, form cues, consultation notes, and draft client message.

Week 5: Client Experience and Analytics

Goal: Client app receives summaries; coach sees analytics dashboard.

1. Client app session recap view with push notifications
2. Send-to-client workflow (coach approves → client notified)
3. Coaching analytics aggregation jobs (talk time, cue patterns, sentiment)
4. Coaching analytics dashboard with visualizations
5. Client sentiment tracking from consultation segments
6. Basic risk detection algorithm (session completion, response time, sentiment trends)

Week 5 Demo: Coach sends session recap → client receives push notification and views personalized summary. Coach views monthly analytics showing talk-time ratio and most common form cues.

Week 6: Polish, Risk Detection, and Launch Prep

Goal: Production-ready MVP with all features working end-to-end.

1. Client risk dashboard with flags and recommended actions
2. Audio cleanup job (24-hour retention policy enforcement)
3. Error handling, loading states, and offline mode polish
4. Performance optimization (database indexing, query tuning, caching)
5. HIPAA compliance audit (encryption verification, audit log review)
6. Production deployment with monitoring (Sentry, performance tracking)
7. Pilot coach onboarding (5-10 coaches for initial feedback)

Week 6 Demo: Full end-to-end demonstration with real pilot coaches. Upload data, record session, generate summary, send to client, view analytics, check risk dashboard. All features operational in production environment.

Workstream Dependencies

Parallel Workstreams: Mobile development and web development can proceed in parallel throughout. Backend API development supports both.

Sequential Dependencies: Week 3 (transcription) must complete before Week 4 (classification/summarization). Week 4 (summaries) must complete before Week 5 (client delivery and analytics).

6. Technical Specifications

6.1 Audio Recording Requirements

Recording Settings

- **Sample Rate:** 16 kHz minimum (optimal for speech-to-text)
- **Bit Depth:** 16-bit
- **Format:** AAC for transmission (smaller files), WAV/PCM for processing
- **Channels:** Mono (single mic)
- **Expected File Size:** ~40MB for 45-minute session (AAC compressed)

Hardware Requirements

The following fitness-grade Bluetooth headset microphones are recommended for optimal transcription accuracy in gym environments:

Microphone	Price	Battery	Notes
Samson Airline 77 Fitness	~\$200	8+ hours	Best value for trainers
SYNCO G2 Max Wireless	~\$150	10+ hours	Budget-friendly option
Shure GLXD14+/SM31	~\$800	12+ hours	Professional grade

6.2 Transcription Pipeline

- **Provider:** Deepgram Nova-2 (batch mode for uploaded recordings)
- **Speaker Diarization:** Enabled (2 speakers: coach and client)
- **Processing Time:** ~1/10th of audio length (45-min session → ~4.5 min)
- **Target Accuracy:** ≥85% with quality headset mic, ≥80% acceptable
- **Retry Policy:** 3 attempts with exponential backoff (60s, 120s, 240s)

6.3 AI Classification and NLP

Intent Classification

Each transcript segment is classified into one of four intents using few-shot prompting with Claude 3.5 Sonnet or GPT-4o:

Intent	Description and Examples
workout_instruction	Exercise commands with sets/reps/weights. "Let's do 3 sets of 10 squats at 135."
form_cue	Technique corrections. "Chest up. Drive through your heels. Knees out."
consultation	Goals, feelings, lifestyle discussion. "How's your sleep been? Any stress this week?"
off_topic	Irrelevant chatter, greetings, gym noise. "Hey, what's up. Nice weather today."

Entity Extraction

From workout_instruction segments, extract: exercise name (normalized), sets, reps, weight value, weight unit. From form_cue segments, extract: cue text, linked exercise, repetition count.

Implementation: Use the Instructor library with Pydantic schemas for structured extraction. Confidence threshold of 70%; below this, flag for manual review.

6.4 Performance Requirements

Operation	Target	Maximum
Audio upload (45-min session, LTE)	30-60 seconds	120 seconds
Transcription processing	4-6 minutes	10 minutes
Classification + extraction	30-45 seconds	60 seconds
Summary generation	10-15 seconds	30 seconds
Total pipeline (end-to-end)	60-90 seconds	180 seconds
Dashboard page load	<500ms	1000ms
API response (simple queries)	<100ms	200ms

6.5 Security and Compliance

- **Encryption at Rest:** AES-256 for all stored data (database, file storage)
- **Encryption in Transit:** TLS 1.2+ for all API communications

- **Authentication:** JWT with short expiry (15 min), refresh tokens with rotation
- **Authorization:** Row-Level Security (RLS) in PostgreSQL; application-level RBAC
- **Audit Logging:** All PHI access logged with user ID, timestamp, action, and entity
- **Data Retention:** Audio files deleted after 24 hours; transcripts retained indefinitely
- **Vendor BAAs:** Required from all vendors handling PHI (Deepgram, cloud storage)

6.6 Error Handling

- **Transcription Failure:** Retry 3x with exponential backoff. After failure, mark session for manual review and notify coach.
- **Low Confidence Classification:** Segments with <70% confidence are flagged with ⚠️ in the summary for coach review.
- **Upload Failure:** Audio cached locally on device. Automatic retry when connection restored. User notified of pending uploads.
- **LLM Timeout:** 30-second timeout per LLM call. On timeout, use simplified summary template without AI elaboration.

7. Acceptance Tests (Demo-Ready Checklist)

Each test represents a specific capability that must work before the MVP is considered demo-ready. Tests are designed to be verifiable with a simple yes/no answer.

7.1 Data Extraction Tests

#	Test Scenario	Pass Criteria
1	Upload InBody PDF	Weight + body fat % appear in profile within 60 sec
2	Upload lab results CSV	Markers populate with values, units, reference ranges, flags
3	Upload Oura/WHOOP export	7/14/30-day HRV and sleep averages displayed

4	Out-of-range lab value	Draft insight card auto-generated in approval queue
5	Coach approval workflow	Draft insight hidden from client until approved

7.2 Session Recording Tests

#	Test Scenario	Pass Criteria
6	Start session recording	Red recording indicator visible, timer counts up
7	Bluetooth headset connection	Audio captured from headset mic (not phone mic)
8	45-minute session recording	Recording completes without crash or audio gaps
9	Background recording (phone locked)	Recording continues when app is backgrounded
10	Upload 45-min session on LTE	Upload completes in <120 seconds

7.3 Transcription and AI Tests

#	Test Scenario	Pass Criteria
11	Speaker diarization	Coach and client utterances correctly labeled (>80%)
12	Exercise extraction	"3 sets of 10 squats at 135" → correct exercise, sets, reps, weight
13	Form cue counting	"Chest up" said 4 times → cue count = 4 in summary
14	Consultation classification	Wedding stress discussion appears in Consultation section, not Workout
15	Summary generation time	Complete summary appears <90 seconds after session end
16	No hallucinations	Summary contains no exercises or details not in transcript

7.4 Client Experience Tests

#	Test Scenario	Pass Criteria
17	Send summary to client	Client receives push notification within 30 seconds
18	Client session recap view	Client sees workout, progress, coach message, homework items

7.5 Analytics and Risk Tests

#	Test Scenario	Pass Criteria
20	Coaching analytics dashboard	Talk-time ratio, cue frequency, sentiment trends displayed
21	Risk score calculation	Client with declining engagement flagged with risk score >70
22	Risk dashboard actions	Recommended actions and conversation starters displayed
23	Audit log completeness	All uploads, approvals, sends logged with timestamp and user

7.6 End-to-End Demo Flow

The following sequence must complete successfully for MVP sign-off:

1. Coach logs in on mobile and web
2. Coach uploads InBody PDF → data extracted and displayed
3. Coach starts session recording with Bluetooth headset
4. Coach conducts 15-minute simulated session (exercises + consultation)
5. Coach taps Complete → summary appears within 90 seconds
6. Summary correctly shows exercises, form cues, consultation notes
7. Coach edits and approves summary
8. Coach sends to client → client receives notification and views recap
9. Coach views analytics dashboard with updated metrics
10. Coach views risk dashboard (if applicable clients exist)

8. Risk Mitigation Strategies

8.1 Technical Risks

Risk 1: Audio Quality in Noisy Gyms

- **Likelihood:** High — gym environments have 80-95 dB background music and equipment noise
- **Impact:** High — poor transcription makes summaries unreliable

- **Prevention:** Require fitness-grade Bluetooth headset mics; test in real gym environments during Week 3; implement noise suppression
- **Fallback:** If accuracy <70%, mark session for manual review; coach can edit summary extensively

Risk 2: Classification Model Inaccuracy

- **Likelihood:** Medium — fitness conversations have domain-specific vocabulary
- **Impact:** Medium — misclassification puts workout data in consultation section (or vice versa)
- **Prevention:** Use few-shot prompting with 20+ carefully curated examples; iterate on prompts with real transcripts
- **Fallback:** Human-in-loop design — coach always reviews before sending; low-confidence segments flagged with ⚠️

Risk 3: LLM Hallucinations

- **Likelihood:** Medium — LLMs can invent plausible-sounding details
- **Impact:** High — invented exercises or advice could harm client trust
- **Prevention:** Strict prompts ("Do not invent details not in transcript"); include raw transcript segments in context; use structured output schemas
- **Fallback:** Coach approval gate — nothing reaches clients without explicit approval

Risk 4: Mobile App Crashes During Recording

- **Likelihood:** Medium — 45-minute background processes stress mobile platforms
- **Impact:** High — lost recording means lost session documentation
- **Prevention:** Incremental audio caching (save every 5 minutes); proper foreground service configuration; extensive testing on multiple devices
- **Fallback:** Recovery mode that uploads partial audio; manual notes entry as backup path

8.2 Product Risks

Risk 5: Coaches Reject Recording Workflow

- **Likelihood:** Medium — some trainers uncomfortable wearing headsets or recording clients
- **Impact:** High — core value proposition depends on session recording
- **Prevention:** Emphasize privacy (audio deleted in 24 hours); clear client consent flow; make time savings visceral ("save 10+ hours/week")
- **Fallback:** Recording is optional feature — coaches can still use data extraction and manual notes without recording

Risk 6: Summary Quality Not "Good Enough"

- **Likelihood:** Medium — initial summaries may feel robotic or miss nuances
- **Impact:** Medium — poor summaries reduce perceived value
- **Prevention:** Extensive prompt iteration during Week 5-6; use pilot coaches for qualitative feedback loops
- **Fallback:** Position as "AI draft, you polish" — expect coach editing; learn coach's style over time

8.3 Business Risks

Risk 7: HIPAA Applicability Uncertainty

- **Likelihood:** Medium — personal trainers exist in regulatory gray area
- **Impact:** High — non-compliance could create legal exposure
- **Prevention:** Consult healthcare attorney in Week 1 to determine applicability; build HIPAA-ready regardless; sign BAAs with all vendors
- **Fallback:** If HIPAA applies, we're already compliant; if not, security posture becomes competitive advantage

Risk 8: Adoption Friction

- **Likelihood:** Medium — trainers may be overwhelmed by new workflow
- **Impact:** High — abandoned after first week means no revenue
- **Prevention:** Phased onboarding (data extraction first, then recording); 30-minute video walkthrough; dedicated Slack support channel
- **Fallback:** Highlight quick wins immediately ("You just saved 18 minutes"); white-glove onboarding for first 50 coaches

9. Definition of Done (MVP Completion)

The MVP is complete when ALL of the following conditions are met. This definition ensures alignment between product, engineering, and stakeholders.

9.1 Feature Completeness

All core features must be functional and tested:

1. **Data Extraction:** InBody PDFs, lab CSVs, and wearable exports correctly extracted and displayed
2. **Insight Generation:** AI-generated insights with coach approval workflow operational
3. **Session Recording:** 45-minute sessions recorded without crashes, with Bluetooth headset support
4. **Transcription:** Audio transcribed with speaker diarization at $\geq 80\%$ accuracy
5. **AI Classification:** Transcript segments classified into workout, cue, consultation, off-topic
6. **Summary Generation:** Structured summaries generated within 90 seconds of session end
7. **Client Delivery:** Summaries delivered to clients via push notification
8. **Coaching Analytics:** Dashboard displaying talk time, cue patterns, sentiment trends
9. **Risk Detection:** At-risk clients flagged with recommended actions

9.2 Quality Requirements

- **No Critical Bugs:** Zero crashes, data loss, or security vulnerabilities in production
- **Performance Targets Met:** All operations within specified latency targets
- **Pilot Validation:** 5+ pilot coaches have successfully recorded real sessions and approved summaries
- **Acceptance Tests Passing:** All 23 acceptance tests from Section 7 pass
- **End-to-End Demo:** Complete demo flow runs without errors for external audience

9.3 Non-Functional Requirements

- **Uptime:** 99.5%+ availability during pilot period
- **Security:** Encryption at rest and in transit; RLS enforced; audit logging active
- **Compliance:** HIPAA audit checklist completed; BAAs signed with all PHI-handling vendors
- **Monitoring:** Error tracking (Sentry) and performance monitoring deployed
- **Documentation:** API documentation, database schema, deployment guide, and coach onboarding materials complete

9.4 Stakeholder Sign-Off

The following individuals must approve MVP completion:

- **CEO:** Confirms product meets strategic vision and is ready for public launch
- **CTO:** Confirms technical architecture is sound and scalable
- **Pilot Coaches:** At least 3 pilot coaches provide written testimonials confirming value

9.5 Launch Readiness

- **Production Environment:** Live and stable with automated deployments
- **Pricing & Billing:** Stripe integration configured for subscription payments
- **Support Channel:** Dedicated support mechanism operational (Slack, email, or in-app)
- **Legal Documents:** Terms of Service, Privacy Policy, and consent forms reviewed by legal counsel
- **App Store Submission:** Mobile app submitted to Apple App Store and Google Play Store (if applicable)

Foundation & Data Processing:

1	Web Chat Interface: Next.js chat UI with OpenRouter integration; model swappable via environment variables
2	Data Intake System: Upload for InBody PDFs, nutrition CSVs, bloodwork, wearables, notes, and audio recordings
3	InBody PDF Extraction: Extract weight_kg and body_fat_pct into Measurements table with units and source tracking
4	Nutrition CSV Processing: Parse nutrition CSV to compute daily averages for protein_g and kcal; store in ProfileFacts

AI Insights & Coach Workflow:

5	Draft Insight Generation: AI-generated InsightCards with title, rationale, actions, confidence; remains draft until approved
6	Coach Approval Workflow: Coach can approve, edit, or reject drafts; only approved insights publish to client plan
7	Async Re-Analysis: Background worker triggers insight generation on new uploads within 60 seconds

AI Notetaker:

8	Live Path (Streaming ASR): In-app audio capture → real-time transcript → LiveTips generation; coach can save tip as insight
9	Fallback Path (Batch Upload): Upload MP3/MP4 → batch ASR → post-call summary → draft insights extraction

Health Data Integrations:

10	Bloodwork Labs: Lab CSV/PDF import with markers (glucose, cholesterol, etc.); out-of-range values generate insights
11	Wearables Data: Daily summary CSVs from Apple Health, Fitbit, Oura, Garmin, WHOOP; 7/14/30-day averages

Platform & Security:

12	Client Profile UI: Consolidated view showing Measurements, ProfileFacts, Labs panel, and Wearables panel
13	Deployment: Next.js deployed; Postgres/Supabase provisioned; OpenRouter configured; queue operational
14	Critical Bug Resolution: No bugs preventing core workflow; end-to-end tests pass
15	HIPAA-First Security: RBAC via Supabase RLS; encryption at rest/transit; PHI-safe logging; immutable audit trail
16	Chat Grounding: Chat responses reference actual extracted measurements; context includes client data

The following tests must **all pass** for MVP Completion. Each test is binary — it either passes or fails.

Core Workflow Tests:

#	Test	Pass Criteria
1	OpenRouter abstraction	Model swap via env variable, no code changes required
2	InBody extraction	Weight and body-fat % stored with units (or N/A with reason)
3	Nutrition CSV	Protein and kcal averages appear in UI and LLM context
4	Chat grounding	Answer cites at least one numerical extracted value
5	Approval gate	Drafts invisible until approved; audit logs editor + timestamp
6	Async re-analysis	New artifact creates draft InsightCard within 60 seconds

AI Notetaker Tests:

#	Test	Pass Criteria
7	Streaming transcript	Live segments persist with accurate timestamps
8	Live tip trigger	Test phrase* creates LiveTip linked to transcript span
9	Tip to insight	Save tip creates draft InsightCard with evidence refs
10	Fallback summary	Uploaded recording creates summary + draft insight

***Test Phrase Clarification:** The "test phrase" is a QA phrase used during acceptance testing — not a wake word like "Hey Siri." During testing, the tester speaks a predetermined phrase (e.g., "Note: client mentioned knee pain during squats") and verifies the system: (1) detects it as a LiveTip trigger, (2) links it to that transcript segment, and (3) allows saving as a draft InsightCard. The exact phrase will be defined during development based on the trigger logic built.

Labs & Wearables Tests:

#	Test	Pass Criteria
11	Lab CSV import	Markers populate with values, units, flags in Labs panel
12	Out-of-range lab insight	Above-range LDL-C produces draft InsightCard
13	Wearable CSV import	Rows normalize into Measurements with units/source
14	Wearable aggregations	7/30-day averages compute and display correctly

Security & Final Tests:

#	Test	Pass Criteria
15	Audit completeness	Upload, approve, publish events logged with user/role/time
16	End-to-end demo	Full scripted flow executes without errors

Phase I:

Highest risk/hardest problems:

1. Data Intake system
 - a. Sample data - sample documents from the coaches
 - b. Challenges - documents have varied formats and the pdf spec is very complex and nuanced.
 - i. Recommendation - use docling, Azure Document Intelligence, AWS Textract, Google Vision,
 1. Extract information
 2. Validate that it was extracted properly - ?? (will we need a manual review step)

- a. Snippets from the original document of where the data came from so that it can be quickly reviewed by a human
 - b. <https://platform.claude.com/docs/en/build-with-claude/citations>
- 2. Trainer Call Audio Processing
 - a. Sample data - sample recordings
 - i. Find YouTube videos of people getting trained by a coach (try to find as much real-life coaching videos as you can)
 - 1. <https://www.youtube.com/shorts/iKyrpNmG1Z8>
 - 2. <https://www.youtube.com/watch?v=5BoOU1tJeFo>
 - b. Real-time streaming - real-time insights
 - c. Post processing analysis - audio to text to insights

Sources of coaching information

- 1. Famous Coaches - Mike Mentzer
- 2. Famous Books - Burn the fat, feed the muscle

We need sample data from real coaches to train/test our AI system.

We can get some data from coaches like Ryan. However, we can also find other sources data from famous coaches -

<https://www.amazon.com/Burn-Fat-Feed-Muscle-Transform/dp/0804137846>

Tom Venuto - Forums - Q/A

One of the hardest problems to solve is getting access to data.

Compute, Algorithms, Data

What is the gold standard?

Corpus of information - we cannot force coaches or clients to confirm a specific format. We have to process data in the format that we receive it.

We may need to develop an approach to build a dataset and it may involve scraping data, downloading youtube videos, joining forums, reading comments/answers on reddit, etc.

There needs to be question/answer

Roles:

- 1. Head of Product
 - a. Create the golden dataset (we need a representative sample of data for all types of data)
 - i. Audio calls
 - ii. Questions/Answers

- iii. Sample lab tests
- iv. Sample inbody scans

Hamzah - find audio samples and create a PoC for processing them

Salman - find data samples and create a PoC for processing them

Top-down - what do you want to accomplish?

Bottom-up - what data can we get access to?

1. Expert data - Ryan Young
2. Open source repositories - kaggle, huggingface, etc
3. Synthetic data using AI - tell gemini or chatgpt to create data
4. Scraping publicly available data (i.e. google, youtube, reddit, x, etc)

Goal #1 - spend a week trying to build a dataset.

https://uk.inbody.com/wp-content/uploads/2018/08/The_Professional_s_Guide_to_the_InBody_Result_Sheet.pdf

InBody scan -

Date

Body fat%

Upper body

Lower body

Classify a document - what type of document is it?

If InBody, extract Date, Body fat%, Upper Body, Lower Body, Weight

Date

Weight

Body fat%

Dexascan

Visceral fat

Body fat

Subcutaneous fat

Quest or LabCorp Blood Tests

Fitness tracker data

Food Journals - MyFitnessPal, LifeSum, etc - calories per day, broken down into macros (i.e. protein, carbs, fat), exercise/calories burned that day

Always do the first things first

- We need to gather the sample data

- We need be able to reliably extract data from the sample data

- AI insights - most compelling portion of the project

Before you can provide the right answer, you have to ask the right questions

We need a database of questions and answers

https://www.reddit.com/r/technology/comments/1ieuxz0/berkeley_researchers_replicate_deepseek_r1_for_30/

Deliverables:

1. Sample Data that you did
2. Analysis of the tools that you tried
 - a. Examples
 - b. Pros/Cons
 - c. Summary/Conclusion - what is your recommendation

Anonymize the name and address (i.e. replace it with John Smith)

Redact the name and address

Remove the PII

If you are going to use production data in development, you need to either anonymize it or redact or get consent by the individual